

Name: _____

Discussion Assignment 9

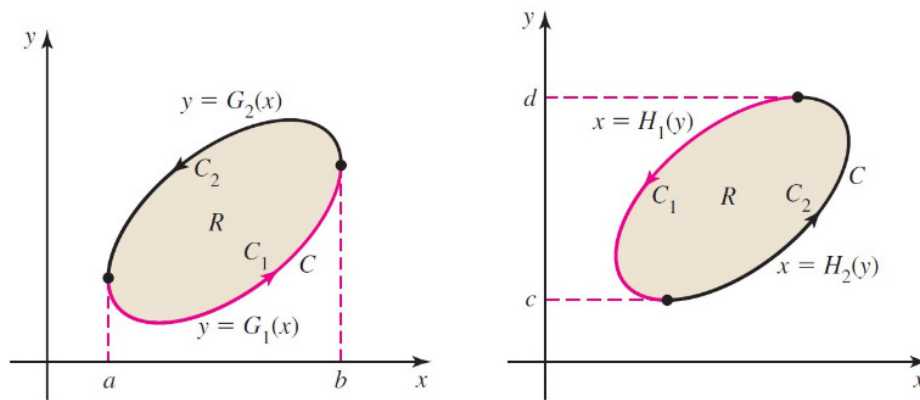
A Green Light at the End of the Tunnel

MATH 2210, Spring 2018

Background

One of the most surprising, beautiful, and unifying conclusions in the study of calculus is the relation between a double integral and the line integral of a vector field. This relationship has many applications in engineering fields, physics, ect. After participating in multivariable calculus for almost an entire semester you are now equipped with enough mathematical skill to be able to write the conclusion for yourself.

Consider the situation below where you have a simple, closed, and smooth curve C that is oriented counter clockwise and defines the boundary of a simply connected region R .



Say your interested in rewriting the integral $\iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$ where f and g are the components of some vector field $\mathbf{F} = \langle f, g \rangle$. You note that you can rewrite the integral $\iint_R \frac{\partial f}{\partial y} dA$ by using the first figure as

$$\iint_R \frac{\partial f}{\partial y} dA = \int_a^b \int_{G_1(x)}^{G_2(x)} \frac{\partial f}{\partial y} dy dx = \int_a^b \left(f(x, G_2(x)) - f(x, G_1(x)) \right) dx.$$

Over the interval $a \leq x \leq b$, the points $(x, G_2(x))$ trace out the upper part of C (labeled C_2) in the negative (clockwise) direction. Similarly, over the interval $a \leq x \leq b$, the points $(x, G_1(x))$ trace out the lower part of C (labeled C_1) in the positive (counterclockwise) direction.

Therefore,

$$\begin{aligned}
 \iint_R \frac{\partial f}{\partial y} \, dA &= \int_a^b \left(f(x, G_2(x)) - f(x, G_1(x)) \right) \, dx \\
 &= \int_{-C_2} f \, dx - \int_{C_1} f \, dx \\
 &= - \int_{C_2} f \, dx - \int_{C_1} f \, dx \\
 &= - \oint_C f \, dx.
 \end{aligned}$$

Question

Employ a similar tactic to rewrite the integral $\iint_R \frac{\partial g}{\partial x} \, dA$.

After combining the results what statement do you end up with?

- (a) $\iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) \, dA = \oint_C f \, dx - g \, dy$
- (b) $\iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) \, dA = \oint_C f \, dx + g \, dy$
- (c) $\iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) \, dA = \oint_C -f \, dx + g \, dy$
- (d) $\iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) \, dA = \oint_C f \, dx - g \, dx$

Individual Work

Now you are to write a report, three page maximum, that convinces an audience of your peers that you have correctly connected the double integral to the line integral of a vector field. This report will need to be saved as a pdf and submitted to your discussion sections WyoCourses page. A scoring rubric and guideline on writing good mathematics is posted on our WyoCourses page.